INCH-POUND

MIL-DTL-32411A
13 December 2019
SUPERSEDING
MIL-DTL-32411
w/AMENDMENT 5
28 August 2015

DETAIL SPECIFICATION

IGNITER, TIME BLASTING FUSE: WITH SHOCK TUBE CAPABILITY - M81 LOAD, ASSEMBLE, AND PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the requirements and verification methods for the following item: IGNITER, TIME BLASTING FUSE: with shock tube capability, M81.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army CCDC, ATTN: FCDD-ACE-SA, Picatinny Arsenal, New Jersey 07806-5000 or e-mailed to usarmy.picatinny.ardec.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil/.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERICAL ITEM DESCRIPTION

A-A-208B	-	Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces)
A-A-59383	-	Marking Media, Opaque (Porous and Non-Porous Surfaces)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-117	-	Bags, Heat-Sealable
MIL-DTL-45144	-	Fuse, Blasting, Time- M700, and Dummy Fuse, Blasting, Time - Inert Loading, Assembly and Packing
MIL-P-20444	-	Primer, Percussion, M42 Parts for and Loading, Assembling and Packing

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	-	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Products
MIL-STD-2073-1	-	Standard Practice for Military Packaging

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

US ARMY COMBAT CAPABILITY DEVELOPMENT COMMAND – ARMAMENTS CENTER (CCDC - AC) DRAWINGS

7553347	-	Box, Wirebound Boxes, Ammunition, M2A2
8822498	-	Cap, Fuse, Holder
8822499	-	Collet
8822500	-	Grommet
8822502	-	Washer, Large
8822503	-	Washer, Small
8822504	-	Firing Assembly
8822505	-	Cap, Top
8822506	-	Housing

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8822507	-	Ring, Pull
8822508	-	Rod, Pull
8822509	-	Spring, Firing Pin
8822510	-	Washer, Friction seal
8822511	-	Washer, Release
8822512	-	Striker Assembly
8822513	-	Housing, Firing Pin
8822514	-	Pin, Firing
9325874	-	Packing For Light Boxes
9345327	-	Primer, Alternative
12972588	-	Base, Primer
12972589	-	Plug, Weatherproofing
12972590	-	Plug, Shipping
12972591	-	Primer, Base Igniter
12972620	-	Igniter, Time, Blasting Fuse: With Shock Tube Capability, M81
12977189	-	Box, Set-Up
12977190	-	Bag, Barrier
12977191	-	Box, Wood
12977192	-	Packing & Marking for Box, Set-Up For Igniter, Time Blasting Fuse w/ Shock Tube Capability M81
12977193	-	Packing & Marking for Box, Wood For Igniter, Time Blasting Fuse w/ Shock Tube Capability M81
13019660	-	Packing and Marking for 54 Igniter, Time Blasting Fuse, Weatherproof, M81
13019661	-	Packing and Marking for 108 Igniter, Time Blasting Fuse, Weatherproof, M81
13026126	-	Shock tube
13032542	-	Box, Ammunition, M2A2 Assembly
13042427	-	Tray
13051505	-	Pin, Cotter

(Copies of the above drawings are available by email at usarmy.pica.ccdc-ac.list.drawing-request-help-desk@mail.mil.)

US ARMY DEFENSE AMMUNITION CENTER (DAC) DRAWINGS

19-48-4116	-	Basic Procedures: Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets
19-48-4116/131B	-	Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets: Igniter, Time Blasting Fuse, M81, Packed 108 per Wirebound Box, Unitized 48 Boxes per 40” x 48” Pallet; Approx Box size 14-7/16” L x 12-17/32” W x 8-1/8” H
19-48-4116/131D	-	Unitization Procedures for Boxed Ammunition and Components on 4-Way Entry Pallets: Igniter, Time

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Blasting Fuse, Packed 360 per Wooden Box, Unitized 24 Boxes per 40" x 48" Pallet; Approx Box size 21-1/2" L x 12" W x 14-1/2" H

(Copies of these documents may be requested online at https://www3.dac.army.mil/DET/order/draworder.html or from U.S. Army Defense Ammunition Center, U.S Amy Technical Center for Explosive Safety, ATTN: ATCL-ACE, 1 C Tree Road, Building 35, McAlester, OK 74501-9053.)

INTERNATIONAL TEST OPERATING PROCEDURES

ITOP 1-2-601	-	Laboratory Vibration Schedules, International Test Operations Procedure

(Copies of the above international test procedures are available online at http://www.atec.army.mil/publications/topsindex.aspx.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B117	-	Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM F519	-	Standard Test Method for Mechanical Hydrogen Embrittlement Evaluation of Coating Processes and Service Environments

(Copies of the above documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. A sample shall be subjected to conformance inspection in accordance with 4.3.

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3.2 Materials, components, and assemblies. All materials, components and assemblies shall comply with the requirements of Drawing 12972620 and its subsidiary drawings.

3.2.1 Hydrogen embrittlement. The firing pin spring and pull ring shall be free of embrittlement.

3.2.2 Salt spray. The firing pin spring, pull ring, and cotter pin shall be subjected to the required salt spray test as specified on the applicable drawing to determine compliance. Except at the edge of an unprotected area, the salt spray tested samples shall show no evidence of corrosion products at the end of the 96 hour test.

3.3 Time fuse initiation. The igniter shall initiate M700 Time Blasting Fuse conforming to MIL-DTL-45144.

3.4 Time fuse attachment integrity. The igniter shall not separate from M700 Time Blasting Fuse, conforming to MIL-DTL-45144, after exposure to not less than 25 pounds (lbs) of axial force for not less than 10 seconds. After this exposure, it shall initiate the time fuse.

3.5 Shock tube initiation. The igniter shall initiate the shock tube conforming to Drawing 13026126.

3.6 Shock tube attachment integrity. The igniter shall not separate from shock tube, conforming to Drawing 13026126, after exposure to not less than 5 pounds (lbs) of axial force for not less than 10 seconds. After this exposure, it shall initiate the shock tube.

3.7 Operating temperatures. The igniter shall operate throughout the temperature range of -50ºF to 130ºF.

3.8 Immersion. The igniter shall operate after immersion in water at a depth not less than 30 inches for not less than 6 hours.

3.9 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.10 Workmanship. All parts, sub-assemblies, and assemblies shall be clean and free of burrs, chips, sharp edges, cracks, unblended radii, burn marks, checks, sinks, crazes, blisters, excess flash, dirt, grease, oil, rust, corrosion, and other foreign matter.

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4. VERIFICATION

TABLE I. Requirement/verification cross reference matrix.

Method of Verification 1 – Analysis 3 – Examination 2 – Demonstration 4 – Test		Classes of Verification A – First article B – Conformance						
Section 3 and 5 Requirements	Description	Verification Method				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article		X	X	X	X		4.2
3.1.2	Conformance inspection		X	X	X		X	4.3
3.2	Materials, components, and assemblies			X	X	X	X	4.3.4.1 – 4.3.4.20
3.2.1	Hydrogen embrittlement				X	X	X	4.5.4
3.2.2	Salt spray				X	X	X	4.5.5
3.3	Time fuse initiation				X	X	X	4.5.6 – 4.5.8
3.4	Time fuse attachment integrity				X	X	X	4.5.6
3.5	Shock tube initiation				X	X	X	4.5.6 – 4.5.8
3.6	Shock tube attachment integrity				X	X	X	4.5.6
3.7	Operating temperatures				X	X	X	4.5.7
3.8	Immersion				X	X	X	4.5.8
3.9	Ammunition lot numbering			X		X	X	4.3.3
3.10	Workmanship			X		X	X	4.3.4.1 – 4.3.4.28
5.1	Packaging and marking			X	X	X	X	4.3.4.21 - 4.3.4.28
5.1.1	Barrier bag resistance to leakage				X	X	X	4.3.4.22 & 4.6.2, 4.6.3
5.1.2	Marking ink/media		X	X		X	X	4.6.5

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows (See Table I):

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When specified, a sample shall be subjected to first article inspection in accordance with Table II. First article samples shall be manufactured using the same vendors, methods, materials, equipment, processes, procedures and facilities as will be used during regular production

4.2.1 First article rejection. If one (1) or more assemblies, subassemblies, materials, or components fail to comply with the first article requirements, the sample(s) and first article shall be rejected.

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TABLE II. First article tests and inspections.

Examination or test	Drawing(s)	Sample Size	Requirement Paragraph	Paragraph Reference / Inspection Method
Ring, pull	8822507			
Examination for defects		25	3.2	4.3.4.1
Hydrogen embrittlement		4.5.4	3.2.1	4.5.4
Salt spray		5	3.2.2	4.5.5
Rod, pull	8822508	25	3.2	4.3.4.2
Spring, firing pin	8822509			
Examination for defects		25	3.2	4.3.4.3
Hydrogen embrittlement		4.5.4	3.2.1	4.5.4
Salt spray		5	3.2.2	4.5.5
Housing, firing pin	8822513	25	3.2	4.3.4.4
Pin, firing	8822514	25	3.2	4.3.4.5
Striker assembly	8822512	25	3.2	4.3.4.6
Housing	8822506	5 1/	3.2	4.3.4.7
Cap, top	8822505	5 1/	3.2	4.3.4.8
Pull rod and striker assembly	8822508 & 8822512	80	3.2	4.3.4.9
Firing assembly	8822504	25	3.2	4.3.4.10
Washer, friction seal	8822510	25 1/	3.2	4/
Washer, release	8822511	25 1/	3.2	4/
Cap, fuse holder	8822498	5 1/	3.2	4.3.4.11
Collet	8822499	5 1/	3.2	4.3.4.12
Plug, weatherproof	12972589	25 1/	3.2	4.3.4.13
Plug, shipping	12972590	5 1/	3.2	4.3.4.14
Base, primer	12972588	25 1/	3.2	4.3.4.15
Primer, alternative	9345327			
Examination for defects		25	3.2	4.3.4.16
Sensitivity test		4.3.4.16	3.2	4.3.4.16
Dud test		4.3.4.16	3.2	
Primer, base igniter	12972591	25 1/	3.2	4.3.4.17
Grommet	8822500	5 1/	3.2	4/
Washer, large	8822502	5 1/	3.2	4/
Washer, small	8822503	5 1/	3.2	4/
Pin, cotter	13051505			
Examination for defects		25	3.2	4.3.4.18
Salt spray		5	3.2.2	4.5.5
M81 Igniter	12972620			
Examination for defects		100 2/	3.2	4.3.4.19

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TABLE III. First article tests and inspections (continued)

Examination or test	Drawing(s)	Sample Size	Requirement Paragraph	Paragraph Reference / Inspection Method
Cotter pin extraction		20 3/	3.2	4.5.1
Pull ring force limits		3/	3.2	4.5.2
Disassembly		20	3.2	8/
Attachment integrities		40	3.4, 3.6	4.5.6
Operating temperatures		160	3.7	4.5.7
Immersion		80	3.8	4.5.8
Packing and marking	12977192	7/	5.1	4.3.4.21 – 4.3.4.28
Setup box	12977189	7/	5.1	4.3.4.21 – 4.3.4.22
Moisture content		4.6.1	5.1	4.6.1
Barrier bags	12977190		5.1	4.3.4.22
Leakage test		5/	5.1.1	4.6.2
Heat seal seam strength test		5/	5.1.1	4.6.3
M2A2 ammunition can (Marine)	13019660	6/	5.1	4.3.4.23 – 4.3.4.24
Marking resistance to water		5.1.2	5.1.2	4.6.5, 4.6.5.1
Marking resistance to smearing		5.1.2	5.1.2	4.6.5, 4.6.5.2
Wood box (Army)	12977193	7/	5.1	4.3.4.25 – 4.3.4.26
Wirebound box (Marine)	13019661	7/	5.1	4.3.4.27
Unitized load		7/	5.1	4.3.4.28

Notes:

1/ If a multiple cavity mold is used, then not less than five (5) samples from each cavity shall be taken. The sample size of the total multiple cavity quantities shall be not less than the quantity specified In Table II. Cavity identification shall be maintained so as to permit tracing each part to the cavity in which it was molded.

2/ After examination for defects, reuse and subject the igniters to FAT tests.

3/ The same igniters shall undergo cotter pin extraction testing followed by pull ring force limits testing.

4/ Verify all dimensions on the drawing.

5/ The test shall be performed on three (3) sealed barrier bags from each sealer.

6/ The tests shall be performed on two (2) marked M2A2 cans (Marine)

7/ For packaging: The examinations shall be performed on one (1) full pack and one (1) light pack

For light packs: The light pack shall be no less than 50% filled and the remainder packed out per Drawing 9325874.

For unitized loads: The examinations shall be performed on one (1) sample.

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Notes (continued):

Packaging within unitized load or assembled packaging may include samples from lower level packaging when conformance can still be demonstrated at all levels. With prior Government approval, the M81s to be enclosed in the packaging may be substituted with inert mass and volume analogues, reused for each sample, or excluded from packaging if not necessary to demonstrate conformance.

8/ Verify all internal components are present, properly assembled and orientated. Accept on zero (0) defects.

4.3 Conformance inspection.

4.3.1 Conformance rejection. If any assembly, subassembly, material, or component fails to comply with the conformance inspection requirements of 4.3 and its subparagraphs, the lot from which the sample was taken shall be rejected.

4.3.2 Inspection lot formation. Inspection lots shall conform to the requirements of MIL-STD-1916. In addition each lot shall contain materials from not more than one lot interfix number from one manufacturer except the following:

a. Primers from not more than one lot from one manufacturer.
b. Primers shall be no older than 5 years from the time of manufacturing.

4.3.3 Ammunition lot number. Visually verify that a unique ammunition lot number has been assigned to each lot as described in MIL-STD-1168.

4.3.4 Inspection requirements by classification of characteristics.

a. For the conformance inspection paragraphs 4.3.4.1 through 4.3.4.29, the definitions of critical, major and minor defects are provided in paragraph “Definitions” of MIL-STD-1916.

b. For the conformance inspection paragraphs 4.3.4.1 through 4.3.4.29, the dimensions listed in the parenthesis “( )” are for reference and or clarification only and may not be the current drawing requirement.

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4.3.4.1 Ring, pull.			Drawing Number 8822507 Next Higher Assembly 8822504	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Hydrogen embrittlement	4.5.4	3.2.1	4.5.4
102	Salt spray	5	3.2.2	4.5.5
Minor				
201	Protective finish damaged, missing, or incomplete	VL II	3.2	Visual
202	Workmanship	VL II	3.10	Visual

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4.3.4.2 Rod, pull.			Drawing Number 8822508 Next Higher Assembly 8822504	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Diameter at front end (.186 +.000/-.002)	VL IV	3.2	Gage
102	Perpendicularity of shoulder with diameter at rear end (.002)	VL IV	3.2	Gage
103	Radius of shoulder (.002 max)	VL IV	3.2	Gage
104	True positon of safety pin hole	VL IV	3.2	Gage
105	Pressure relief cavity or hole incomplete or blocked	VL IV	3.2	Visual
106	Diameter of cotter pin hole (.078 +.004/- .001)	VL IV	3.2	Gage
107	Diameter of relief cavity (.078 +.004/-.001)	VL IV	3.2	Gage
108	Depth of relief cavity(1.000 +/-.010)	VL IV	3.2	Gage
109	Diameter of relief hole (.078 +.004/-.001)	VL IV	3.2	Gage
110	True position of relief hole	VL IV	3.2	Gage
111	Small diameter of rod (.149 +.000/ -.003)	VL IV	3.2	Gage
Minor				
201	Total length (2.17 +.00/-.02)	VL II	3.2	Gage
202	Finish damaged, missing, or incomplete	VL II	3.2	Visual
203	Workmanship	VL II	3.10	Visual

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4.3.4.3 Spring, firing pin.			Drawing Number 8822509 Next Higher Assembly 8822504	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Hydrogen embrittlement	4.5.4	3.2.1	4.5.4
102	Solid height (.84 max)	VL IV	3.2	Gage
103	Load at .89” height (4.2 +/-.4 lbs)	VL IV	3.2	Gage
104	Protective finish damaged, missing, or incomplete	VL IV	3.2	Visual
105	End not squared or ground	VL IV	3.2	Visual
106	Salt spray	5	3.2.2	4.5.5
Minor				
201	Workmanship	VL II	3.10	Visual

4.3.4.4 Housing, firing pin.			Drawing Number 8822513 Next Higher Assembly 8822512	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Perpendicularity of cavity base with inside diameter at front end (.002)	VL IV	3.2	Gage
102	Outside diameter (.219+.000/-.003)	VL IV	3.2	Gage
103	Inside diameter at front end (.187 +.002/-.000)	VL IV	3.2	Gage
104	Total length (.89 +.00/-.01)	VL IV	3.2	Gage
Minor				
201	Finish damaged, missing, or incomplete	VL II	3.2	Visual
202	Presence of slots (4x)	VL II	3.2	Visual
203	Workmanship	VL II	3.10	Visual

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4.3.4.5 Pin, firing.			Drawing Number 8822514 Next Higher Assembly 8822512	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Total length (.45 +.00/-.01)	VL IV	3.2	Gage
102	Length of point (.100 +.015/-.000)	VL IV	3.2	Gage
103	Profile of point (spherical .08 +.00/-.01)	VL IV	3.2	Gage
104	Location of crimp groove (.13 +00/-.01)	VL IV	3.2	Gage
105	Hole diameter (.054 +.005/-.000, x10)	VL IV	3.2	Gage
106	.110 min from center of part to inside edge of hole (x10)	VL IV	3.2	Gage / Visual
107	Outside diameter not breached or compromised by holes	VL IV	3.2	Visual
108	True position of flange diameter	VL IV	3.2	Gage
Minor				
201	Distance from base to flange (.25 +.01/-.00)	VL II	3.2	Gage
202	Workmanship	VL II	3.10	Visual

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4.3.4.6 Striker assembly.			Drawing Number 8822512 Next Higher Assembly 8822504	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Perpendicularity of forward face of flange with outside diameter of firing pin housing (.003)	VL IV	3.2	Gage
102	Total length (1.10 max)	VL IV	3.2	Gage
Minor				
201	Crimp improper (shallow or less than 360°)	VL II	3.2	Visual
202	Holes not obstructed	VL II	3.2	Visual
203	Workmanship	VL II	3.10	Visual

4.3.4.7 Housing.			Drawing Number 8822506 Next Higher Assembly 8822504	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Major diameter of thread (5/8-20 UN-2A, x2)	VL IV	3.2	Gage
102	Pitch diameter of thread (5/8-20 UN-2A, x2)	VL IV	3.2	Gage
103	Inside diameter (.39 +.01/-.00)	VL IV	3.2	Gage
104	Total length (3.18 +.00/-.03)	VL IV	3.2	Gage
105	Threads damaged	VL IV	3.2	Visual
Minor				
201	Outside diameter (.627 +.000/-.010)	VL II	3.2	Gage
202	Perpendicularity of closed end surface	VL II	3.2	Gage
203	Diameter of pull rod hole (.150 +.005/-.000)	VL II	3.2	Gage
204	Workmanship	VL II	3.10	Visual

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4.3.4.8 Cap, top.			Drawing Number 8822505 Next Higher Assembly 8822504	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Distance from front end to bottom of thread (.56 +.05/-.00)	VL IV	3.2	Gage
102	Total length (.95 +.00/-.02)	VL IV	3.2	Gage
103	Pitch diameter of thread (5/8-20 UN-2B)	VL IV	3.2	Gage
104	Minor diameter of thread (5/8-20 UN-2B)	VL IV	3.2	Gage
105	Location of cotter pin hole (.80 +/-.01)	VL IV	3.2	Gage
106	Crack or split	VL IV	3.2	Gage
107	Surface in bottom not smooth or free of flash	VL IV	3.2	Visual
Minor				
201	Diameter of pull rod hole (.150 +.005/-.000)	VL II	3.2	Gage
202	Workmanship	VL II	3.10	Visual

4.3.4.9 Subassembly of pull rod and striker assembly.			Drawing Number 8822508 & 8822512 Next Higher Assembly 8822504	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Firing pin load test	1/	3.2	4.5.3
Minor				
201	Workmanship	1/	3.10	Visual

Notes:
1/ Test 50 subassemblies of the pull rods and striker assemblies per igniter lot.

If one (1) or more subassemblies fail to comply with the specified requirement, then the M81, pull rod, firing pin, and striker assembly lots shall be rejected.

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4.3.4.10 Firing assembly.			Drawing Number 8822504 Next Higher Assembly 12972620	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Pull rod not engaged with striker assembly	VL IV	3.2	Manual
102	Depth from fuse holder end to firing pin flange (1.12 +.06/-.00)	VL IV	3.2	Gage
103	External components missing or loose (pull ring, top cap, cotter pin with cord, pull rod, striker assembly)	VL IV	3.2	Visual
Minor				
201	Cord improperly tied	VL II	3.2	Visual
202	Threads damaged	VL II	3.2	Visual
203	Workmanship	VL II	3.10	Visual

4.3.4.11 Cap, fuse holder.			Drawing Number 8822498 Next Higher Assembly 12972620	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Pitch diameter of thread (5/8-20 UN-2B)	VL IV	3.2	Gage
102	Minor diameter of thread (5/8-20 UN-2B)	VL IV	3.2	Gage
103	Total length (1.10 +.00/-.03)	VL IV	3.2	Gage
104	Distance from rear end to cavity below thread (.78 +.02/-.00)	VL IV	3.2	Gage
105	Distance from rear end to bottom of thread (.63 +.05/-.00)	VL IV	3.2	Gage
Minor				
201	Workmanship	VL II	3.10	Visual

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4.3.4.12 Collet.			Drawing Number 8822499 Next Higher Assembly 12972620	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Inside radius (.095 +.005/-.000)	VL IV	3.2	Gage
Minor				
201	Workmanship	VL II	3.10	Visual

4.3.4.13 Plug, weatherproofing.			Drawing Number 12972589 Next Higher Assembly 12972620	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Outside diameter of smallest diameter (.118 +.000/-.005)	VL IV	3.2	Gage
102	Length of smallest diameter (1.00 +.00/-.03)	VL IV	3.2	Gage
Minor				
201	Overall length (1.61 +.00/-.03)	VL II	3.2	Gage
202	Workmanship	VL II	3.10	Visual

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4.3.4.14 Plug, shipping.			Drawing Number 12972590	
			Next Higher Assembly 12972620	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Inside diameter (.118 +.005/-.000)	VL IV	3.2	Gage
102	Overall length (1.00 +.00/-.03)	VL IV	3.2	Gage
103	Outside diameter of small outside diameter (.212 +.000/-.005)	VL IV	3.2	Gage
Minor				
201	Outside diameter of knob (.625 +.000/-.020)	VL II	3.2	Gage
202	Thickness of knob (.12 +.00/-.02)	VL II	3.2	Gage
203	Workmanship	VL II	3.10	Visual

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4.3.4.15 Primer, base.			Drawing Number 12972588 Next Higher Assembly 12972591	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Overall length (.51 +.00/-.01)	VL IV	3.2	Gage
102	Inside diameter of small diameter hole (.100 +.005/-.000)	VL IV	3.2	Gage
103	Inside diameter of counter bore (.1715 +.0030/-.0000)	VL IV	3.2	Gage
104	Length of shoulder from slotted end (.130 +.000/-.005)	VL IV	3.2	Gage
105	Depth of counter bore (.116 +.003/-.000)	VL IV	3.2	Gage
106	Outside diameter of large diameter (.445 +.000/-.005)	VL IV	3.2	Gage
107	Outside diameter of small diameter (.373 +.000/-.005)	VL IV	3.2	Gage
108	True position of 5 longitudinal slots relative to alignment	VL IV	3.2	Gage
109	Depth of 5 longitudinal slots (.060 +.005/-.000)	VL IV	3.2	Gage
110	True position of longitudinal alignment key	VL IV	3.2	Gage
Minor				
201	Workmanship	VL II	3.10	Visual

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4.3.4.16 Primer, alternative.			Drawing Number 9345327 Next Higher Assembly 12972591	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Height (.115 +/- .119)	VL IV	3.2	Gage
102	Diameter (.1752 +/- .0004)	VL IV	3.2	Gage
103	Sensitivity testing	1/	3.2	1/
104	Dud testing	2/	3.2	2/
Minor				
201	Workmanship	VL II	3.10	Visual
Notes: 1/ Sensitivity testing shall be in accordance with MIL-P-20444 paragraph 4.2.3.2.1 and 9345327. 2/ Dud testing shall be in accordance with MIL-P-20444 paragraph 4.2.3.2.2 and 9345327.				

4.3.4.17 Primer, base igniter subassembly.			Drawing Number 12972591 Next Higher Assembly 12972620	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Primer missing, inverted, or damaged	VL IV	3.2	Visual
102	Primer not fully seated	VL IV	3.2	Visual
Minor				
201	Workmanship	VL II	3.10	Visual

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4.3.4.18 Pin, cotter.			Drawing Number 13051505 Next Higher Assembly 8822504	
	Examinations and Tests	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Salt spray	5	3.2.2	4.5.5
Minor				
201	Protective finish damaged, missing, or incomplete	VL II	3.2	Visual
202	Workmanship	VL II	3.10	Visual

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4.3.4.19 Igniter, time, blasting fuse/shock tube capability – examinations.			Drawing Number 12972620 Next Higher Assembly	
	Examinations and Tests	Conformance Criteria 1/	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Total length (5.343 max.)	VL IV	3.2	Gage
102	Length from top cap to end of fuse holder cap (4.00 max.)	VL IV	3.2	Gage
103	External components missing or loose (top cap, pull rod, weather proof plug, shipping plug, fuse holder cap, cotter pin, pull ring, cord)	VL IV	3.2	Visual/ Manual
104	Top cap, fuse holder cap or housing cracked or split	VL IV	3.2	Visual
105	Markings missing, illegible or incorrect	VL IV	5.1.2	4.6.5
Minor				
201	Cotter pin not spread equally	VL II	3.2	Visual
202	Cotter pin cord missing or improperly tied	VL II	3.2	Visual
203	Protective finish missing, or incomplete (pull ring and cotter pin)	VL II	3.2	Visual
204	Marking color improper	VL II	3.2	Visual
205	Knurl missing (where applicable)	VL II	3.2	Visual
206	Workmanship	VL II	3.10	Visual

Notes:
1/ Sample size shall be in accordance with the MIL-STD-1916 verification level, plus all the destructive test units of 4.3.4.20.

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4.3.4.20 Igniter, time, blasting fuse/shock tube capability–destructive.			Drawing Number 12972620 Next Higher Assembly	
	Examinations and Tests	Conformance Criteria 1/	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Cotter pin extraction	20 2/	3.2	4.5.1
102	Pull ring force limits	2/	3.2	4.5.2
103	Attachment integrities	80	3.4, 3.6	4.5.6
104	Operating temperatures	100	3.7	4.5.7
105	Immersion	100	3.8	4.5.8
106	Disassembly	20	3.2	3/

Notes:

1/ The total function test quantity is 320.

The functional tests results shall be summed together and the following rejection criteria used:

During testing, if two (2) or more assemblies fail to initiate after the third attempt, the lot shall be rejected.

Any test failure(s) caused by either improper assembly of the igniter or inability to be reset qualifies both as a test and lot failure.

2/ The same sample igniters shall undergo sequential cotter pin extraction testing followed by pull ring force limits testing.

The tests results shall be combined and the following acceptance/rejection criteria used:

Accept on zero (0) defects; reject on one (1) or more defects for any force less than each respective force requirement.

Accept on one (1) defect; reject on two (2) or more defects for any force greater than each respective force requirement.

3/ Verify all internal components are present, properly assembled and orientated.

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4.3.4.21 Packing and marking for box, set-up for M81 igniter (prior to sealing).			Drawing Number 12977192 Next Higher Assembly 13019660 19277193	
	Examination	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Missing items	VL II	5.1	Visual
102	Packaging material moisture content incorrect	4.6.1	5.1	4.6.1
Minor				
201	Workmanship	VL I	3.10	Visual

4.3.4.22 Packing and marking for box, set-up for M81 igniter (after sealing).			Drawing Number 12977192 Next Higher Assembly 13019660 19277193	
	Examination	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Sealing strip torn, badly wrinkled or otherwise fails to seal box completely or correctly	VL II	5.1	Visual
102	Markings missing, incorrect, illegible, incorrect size/height, or not in proper location	VL II	5.1.2	4.6.5 / Gage
103	Leakage test	1/	5.1.1	4.6.2
104	Heat seal seam strength test	1/	5.1.1	4.6.3
Minor				
201	Contents loose	VL I	5.1	Manual
202	Workmanship	VL I	3.10	Visual

Notes

1/ The test shall be performed at the beginning of each shift on one (1) sealed barrier bag from each sealer. For sampling purposes, consider a shift to be no more than 12 hours.

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4.3.4.23 Packing and marking drawing for the M81 Marine packaging M2A2 ammo can (prior to sealing).			Drawing Number 13019660 Next Higher Assembly 13019661	
	Examination	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Missing items	VL II	5.1	Visual
102	Incorrect humidity indicator card or desiccant bag	VL II	5.1	Visual
Minor				
201	Workmanship	VL I	3.10	Visual

4.3.4.24 Packing and marking for the M81 Marine M2A2 ammo can (after sealing).			Drawing Number 13019660 Next Higher Assembly 13019661	
	Examination	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Metallic seal missing, missing manufacturer's identification, incorrect, not secured, or not applied correctly	VL II	5.1	4.6.4
102	Markings missing, incorrect, illegible, incorrect size/height, or not in proper location	VL II	5.1.2	4.6.5/ Gage
103	Can punctured, dented, cracked, or damaged	VL I	5.1	Visual
Minor				
201	Contents loose	VL I	5.1	Manual
202	Workmanship	VL I	3.10	Visual

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4.3.4.25 Packing and marking drawing for the M81 Army wood box (prior to sealing).			Drawing Number 12977193 Next Higher Assembly	
	Examination	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Missing items	VL II	5.1	Visual
102	Markings – Preservative, Heat Treatment, and Box Manufacturer – missing, incorrect, illegible, incorrect size/height, or not in proper location	VL II	5.1.2	Visual/ Gage
Minor				
201	Workmanship	VL I	3.10	Visual

4.3.4.26 Packing and marking drawing for the M81 Army wood box (after sealing).			Drawing Number 12977193 Next Higher Assembly 19-48-4116/131D (M81 Army)	
	Examination	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Box, hinge, or hasp damaged	VL II	5.1	Visual
102	Metallic seal(s) missing, missing manufacturer's identification, incorrect, not secured, or not applied correctly	VL II	5.1	4.6.4
103	Markings –including 2D bar codes - missing, incorrect, illegible, incorrect size/height, or not in proper location	VL II	5.1.2	4.6.5/ Gage
Minor				
201	Contents loose	VL I	5.1	Manual
202	Workmanship	VL I	3.10	Visual

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4.3.4.27 Packing and marking drawing for the M81 Marine wirebound box overpack (prior to sealing).			Drawing Number 13019661 Next Higher Assembly N/A	
	Examination	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Missing Items	VL II	5.1	Visual
102	Markings – Preservative, Heat Treatment, and Box Manufacturer- missing, incorrect, illegible, incorrect size/height, or not in proper location	VL II	5.1.2	Visual/ Gage
Minor				
201	Workmanship	VL I	3.10	Visual

4.3.4.28 Packing and marking drawing for the M81 Marine wirebound box overpack (after sealing).			Drawing Number 13019661 Next Higher Assembly 19-48-4116/131B (M81 Marine)	
	Examination	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Box, hinge, or hasp damaged	VL II	5.1	Visual
102	Metallic seal(s) missing, missing manufacturer's identification, incorrect, not secured, or not applied correctly	VL II	5.1	4.6.4
103	Markings –including 2D bar codes - missing, incorrect, illegible, incorrect size/height, or not in proper location	VL II	5.1.2	4.6.5/ Gage
Minor				
201	Contents loose	VL I	5.1	Manual
202	Workmanship	VL I	3.10	Visual

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4.3.4.29 Unitization.			Drawing Number 19-48-4116/131B (M81 Marine) 19-48-4116/131D (M81 Army)	
			Next Higher Assembly	
	Examination	Conformance Criteria	Requirement Paragraph	Paragraph Reference/ Inspection Method
Critical	None defined			
Major				
101	Wood preservative and heat treatment markings for the pallet missing, incorrect, illegible, or not in proper location	VL II	5.1	Visual
102	Pallet load markings -including 2D bar codes - missing, incorrect, illegible, or not in proper location	VL II	5.1.2	4.6.5
103	Filler assembly not built, treated, or marked correctly	VL II	5.1	Visual
104	Boxes incorrectly oriented or positioned on pallet	VL II	5.1	Visual
105	Strapping not applied correctly or incorrect type	VL II	5.1	Visual
106	Unit load not tightly strapped or any strap loose	VL II	5.1	Visual / Manual
Minor				
201	Workmanship	VL I	3.10	Visual

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4.4 Test sub-procedures.

4.4.1 M700 insertion procedures. Loosen the fuse holder cap three revolutions. Loosen the plugs by pressing them into the igniter. Grip both the weatherproofing and shipping plugs and move them side to side while simultaneously pulling both plugs from the igniter. Cut a 24" ±1" length of fuse so its ends are clean and square. Insert the fuse into the igniter as far as it will go (to ensure complete and proper insertion of the fuse). Re-tighten the fuse holder cap to insure proper holding force; fingertight closures shall be used.

4.4.2 Shock tube insertion procedures. Loosen the fuse holder cap three revolutions. Loosen the weatherproofing plug (outermost plug) by pressing the plug into the igniter. Grip the weatherproofing plug and move it side to side while simultaneously pulling the weatherproofing plug out of the shipping plug and the igniter. Cut a 24" ±1" length of shock tube so its ends are clean and square. Seal one end of the shock tube (tape, ultrasonic, cap, etc.). Insert the unsealed end of the shock tube into the shipping plug as far as it will go (to ensure complete and proper insertion of the shock tube). Re-tighten the fuse holder caps to insure proper holding force; fingertight closures shall be used.

4.4.3 Functioning procedures. Secure the igniter. Remove the cotter pin. Apply an axial force to the pull ring, within the limits specified on Drawing 12972620. If the M81 does not function, reset per 4.4.3.1. The M81 and lot shall be considered a failure if the attached time fuse or shock tube length does not initiate.

4.4.3.1 Quick reset procedures. If the igniter does not function on the first initiation attempt, reset the igniter. To reset the igniter, reengage the striker assembly by pushing the pull rod back into the end of the igniter. Verify the pull rod re-engages the striker by feeling the pull ring's increased resistance to withdrawal. After the pull rod re-engages the striker, apply an axial force to the pull ring, within the limits specified on Drawing 12972620. If the igniter does not function after the first reset, it may be reset a second time. The M81 shall be rejected if the igniter is incapable of being reset or if it is reset twice and still does not function.

4.5 Test methods.

4.5.1 Cotter pin extraction test. This test shall be performed on the M81. Secure the igniter. Beginning at not greater than 9 lbs, apply the incremental axial force to the eye of the cotter pin in the direction away from the igniter and cotter pin until it withdraws completely. The M81 and lot shall be rejected if the cotter pin is withdrawn from the igniter at less than 10 lbs or greater than 30 lbs is required for withdrawal.

4.5.2 Pull ring force limits test. This test shall be conducted on the M81. Secure the igniter. Remove the cotter pin. Beginning at not greater than 4 lbs, apply an incremental axial force to the pull ring in the direction away from the igniter until it functions (as evidenced by audible noise). The M81 and lot shall be rejected if it functions at less than 5 lbs or greater than 15 lbs is required for functioning.

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4.5.3 Firing pin load test. This test shall be performed on a subassembly consisting of the pull rod assembled to the striker assembly, conforming to Drawings 8822508 and 8822512 respectively. Secure the pull ring end of the pull rod. Apply an axial load not less than 15 lbs to the firing pin flange in a direction away from the pull rod. The striker assembly and pull rod and their lots shall be rejected if the roll crimp fails or the pull rod disengages.

4.5.4 Hydrogen embrittlement. This test shall be performed initially and no less than every 120 days to assure quality and process control within the zinc/chromate plating process utilized during manufacture of the firing pin spring and pull ring. Perform the test in accordance with ASTM F519 using the specified number of Type 1a.1 Standard size test specimens. If one (1) or more test fails, the parts produced using the plating/coating process since the last successful test are considered embrittling and shall be rejected.

4.5.5 Salt spray. This test shall be conducted on the pull ring, firing pin spring, and cotter pin to determine compliance with the applicable drawings' requirements. The test shall be performed in accordance with ASTM B117 and the test duration shall be 96 hours, per the drawings and referenced specifications. The frequency of the test shall be once per week. If multiple parts use the same plating process, the test need not be performed specifically during a pull ring, firing pin spring, or cotter pin plating batch; rather vary the testing batch to check the plating process across all parts. If one (1) or more samples have corrosion products (visible to the unaided eye at normal reading distance) the parts produced using the plating/coating process since the last successful test shall be rejected.

4.5.6 Attachment integrity test. This test shall be conducted on the M81. Equally divide the M81 samples into the two groups, one for insertion with time fuse and one for insertion with shock tube. Obtain lengths of M700, conforming to MIL-DTL-45144 and shock tube, conforming to Drawing 13026126. Insert the lengths into M81s per 4.4.1/4.4.2. Secure the igniter and apply the specified axial force for the specified time to the junction of the M81 and length. The M81 shall be rejected if the length separates from the igniter. Carefully unsecure the igniter from the fixture. Seal the shock tube ends (tape, ultrasonic, cap, etc.). The M700 lengths may be cut to no less than a 3 inch protrusion from the igniter. Initiate the igniters per 4.4.3. Visually inspect the M700 for signs of initiation (venting, deformation, flame exiting the end). Visually inspect the shock tube for breakage of the end seal (and lack of interior explosive powder if the end seal is not broken). The M81 and lot shall be rejected if it does not initiate the time fuse/shock tube.

4.5.7 Operating temperatures. Equally divide the igniters into the two temperature groups. The items tested at Hot shall be conditioned at 125ºF +5/-0 ºF. The items tested at Cold shall be conditioned at -35ºF +0/-15ºF. Each of the two groups shall be conditioned for a not less than of 4 hours at their respective temperature. Along with the igniters, condition time fuse and shock tube, conforming to MIL-DTL-45144 and Drawing 13026126 respectively. From each temperature group, half of the samples shall be used to initiate time fuse and the other half shall initiate shock tube. Within 5 minutes after the removal from the chamber, the igniter shall be inspected and functioned. Insert similarly conditioned time fuse/shock tube per 4.4.1/4.4.2. The M700 lengths may be cut to no less than a 3 inch protrusion from the igniter. Initiate the igniters per 4.4.3. Visually inspect the time fuse and shock tube for signs of initiation (venting,

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deformation, flame exiting the end, etc.). The M81 and lot shall be rejected if it does not initiate the time fuse /shock tube.

4.5.8 Immersion test. The test shall be conducted on the M81. Equally divide the M81 samples into the two groups, one for immersion with time fuse and one for immersion with shock tube. Obtain lengths of M700 and shock tube, conforming to MIL-DTL-45144 and Drawing 13026126 respectively, and attach igniters onto both ends of the lengths per 4.4.1/4.4.2. Then submerge the entire length and attached igniters into the specified depth of water for the specified length of time. Remove the lengths and igniters from the water and within 30 minutes of removal from the water, cut the lengths midway between the igniters. The M700 lengths may be cut to no less than a 3 inch protrusion from the igniter. Seal the shock tube ends (tape, ultrasonic, cap, etc.). Initiate the igniters per 4.4.3. Visually inspect the M700 for signs of initiation (venting, deformation, flame exiting the end, etc.). Visually inspect the shock tube for breakage of the end seal (and lack of interior explosive powder if the end seal is not broken). The M81 and lot shall be rejected if it does not initiate the time fuse/shock tube.

4.6 Packaging inspections.

4.6.1 Packaging material moisture content. At the start of each production day, three (3) samples of all paper packaging material contained within the barrier bag (setup box) shall be randomly selected and subjected to moisture content analysis. No less than ten (10) grams of each sample shall be accurately weighed and placed in a tared weighing dish. Each dish and contents shall be weighed and placed in an oven and dried at 100± 2°C for 2 hours. Each dish and contents shall be cooled in a desiccator for a minimum of 15 minutes and weighed. The loss in weight shall be calculated as percent moisture in each sample. If any sample does not meet the moisture requirement, the sample(s) and that production day's packaging material shall be rejected.

4.6.2 Leakage test. The test shall be performed on the sealed barrier bag and its contents (actual or simulated) in accordance with MIL-STD-2073-1 Leakage Test paragraph G.4.2. The test shall follow either the vacuum retention, the submersion (or immersion), or the hot water technique.

4.6.3 Heat seal seam strength test. The test shall be performed on the sealed barrier bag in accordance with MIL-STD-2073-1 Heat seal seam strength test paragraph G.4.3. The samples shall be chosen from the final heat seal seam which closes/seals the pre-assembled bag.

4.6.4 Metallic seal inspection. The metallic seal shall be inspected for presence, proper markings, and installed correctly. Pull on the metallic seal to determine if it is secure. If the seal breaks, the sample and item lot shall be rejected. If the wire breaks before the seal, the box shall be resealed using a new metallic seal.

4.6.5 Markings. Verify by visual inspection. The sample and lot shall be rejected if it does not conform to the marking requirements.

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4.6.5.1 Marking resistance to water. The test shall be performed on First Article marked M2A2 ammunition can samples in accordance with the water resistance performance criteria of its respective drawing or specification (A-A-59383 paragraph 3.6 and A-A-208B paragraph 5.5, as applicable). The packaging material used shall be from lots submitted as part of First Article. The sample(s) and lot shall be rejected if the marking media is not legible, retains its color, smears, blisters, cracks, or peels. (Note: This testing is in addition to any testing required for the ink certification.)

4.6.5.2 Marking resistance to rubbing. The test shall be performed on marked M2A2 ammunition can samples in accordance with the smearing resistance performance criteria of its respective drawing or specification (A-A-59383 paragraph 3.7 and A-A-208B paragraph 5.4, as applicable). The packaging material used shall be from lots submitted as part of First Article. The sample(s) and lot shall be rejected if the marking media smears. (Note: This testing is in addition to any testing required for the ink certification.)

5. PACKAGING

5.1 Packaging and marking. The materials, components, and assemblies shall comply with all the requirements of the drawings listed in Table III and all its subsidiary drawings.

TABLE III. Applicable M81 packaging drawings.

ITEM	PACKING AND MARKING DWG(S)	SET-UP BOX DWG	BARRIER BAG DWG	OUTER PACK DWG	OVER PACK DWG	MISCELLANEOUS PACKAGING MATERIAL DWG(S)	BASIC UNITIZATION DWG 19-48-4116 AND APPENDIX
M81 (Army)	12977192/ 12977193	12977189	12977190	12977191	N/A	N/A	19-48-4116/ 131D
M81 (Marine)	12977192/ 13019660/ 13019661	12977189	12977190	13032542	7553347	13042427	19-48-4116/ 131B

5.1.1 Barrier bag resistance to leakage. The barrier bag shall not leak as evidenced by performing MIL-STD-2073-1 Para G.4.2 (leakage test) and MIL-STD-2073-1 Para G.4.3 (heat-sealed seam test).

5.1.2 Marking ink/media. The ink and/or marking media shall be in accordance with A-A-208B, A-A-59383, or other specified specification. If a quantity is not specified in the specification, a sample of three (3) per ink/paint/surface/condition to be marked shall be used. All markings shall be smooth, uniform, clear in appearance and free of conspicuously ragged edges. There shall be no bleeding of color markings or the plastic exterior to the extent that there is a mottled or streaky appearance. The markings shall show no lifting, wrinkling, or other visible effect on the plastic's surface. The markings shall be applied on clean, dry surfaces that shall be free of dirt, oil, grease, water, corrosion, or other foreign matter. All required markings shall be present, neat, legible, only where specified, and sharply defined

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6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The M81 is a weatherproof igniter intended to be used for the initiation of either time blasting fuse or shock tube. The M81 is part of the Modernized Demolition Initiators (MDI) family. The M81 is used to initiate all past and current fielded and developmental time blasting fuses and shock tubes. The M81, covered by this specification, is intended for use with military demolitions. These M81s, covered by this specification, are military unique as they should function under rough handling and extreme climatic and storage conditions as encountered by the United States Armed Forces.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, and of all reference documents cited in Section 2 and applicable documents from Section 6.
b. Requirement/CDRL for Physical Configuration Audit (PCA).
c. Requirement/CDRL for First article inspection.
d. Requirement/CDRL for conformance inspection.
e. Requirement/CDRL for Ammunition Data Cards (ADC) as specified in MIL-STD-1168.
f. Requirement/CDRL for Acceptance Inspection Equipment (AIE) and associated test procedures. M700 and shock tube used during LAT should be submitted as part of the AIE.
g. Requirement/CDRL for Alternate Acceptance Plans.
h. Packaging requirements. Special requirements exist for the packaging of this item. Detailed directions and additional information may be obtained from FCDD-ACE-LP.
i. Requirement for certificates of findings (certificates of analysis) on the primer lot(s) to be submitted with each igniter lot.
j. Requirement for evidence that the primer is manufactured by a qualified source and the total time between original acceptance of the primer lot and its assembly into the igniter should not be greater than 2 years.

6.3 Complete functioning. The functioning of the M81 can be evidenced by the propagation of the M81's primer to the inserted time fuse or shock tube. The time fuse can be verified for propagation by looking for a flaming exiting the time fuse end and/or by examining the exterior and interior, which changes in appearance when burned. The shock tube can be verified for propagation by looking for a flame exiting the shock tube and/or examining the interior walls for explosive material. If the shock tube's end is sealed (by tape, ultrasonic, cap, etc.) then a rupture of the shock tube or seal will typically occur during functioning. If the seal or shock tube does not rupture, the shock tube can be cut open and the interior walls inspected for the absence of explosive.

6.4 Document history. This document supersedes the purchase description QAA-1422E Item Specification; Igniter, Time Blasting Fuse with Shock Tube Capability: M81, Parts for, and Loading, Assembling and Packing; dated 25 April 2008.

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6.4.1 Amendment overview of changes. The following approved Engineering Change Proposals (ECPs) are being incorporated into this document: R19B2001 and R16Q2017.

6.5 Subject term (key word) listing.
Friction seals
Modernized Demolition Initiators (MDI)
Pull ring

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AR
Navy – MC
Air Force - 170

Preparing activity:
Army-AR
(Project 1375-2017-003)

Review activities:
Navy – AS, SH
Air Force – 11, 20, 70

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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